AMOVLAB
一阿木实验室一

Human-like navigation in human-living environments

Tingxiang Fan

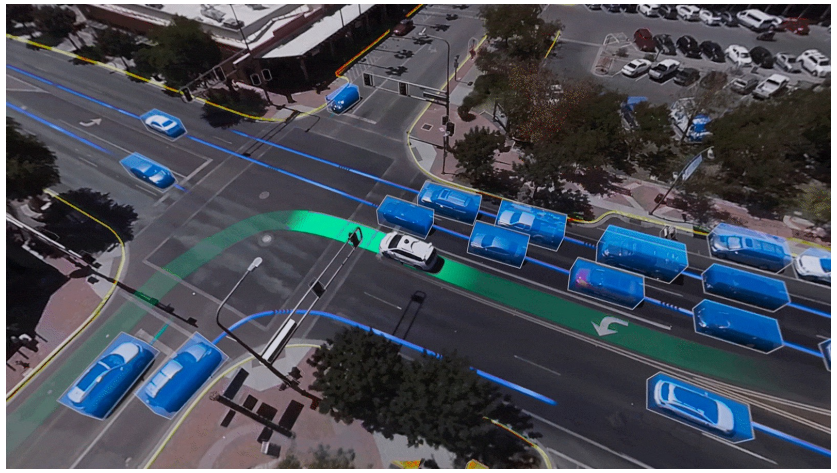


THE UNIVERSITY OF HONG KONG
DEPARTMENT OF
COMPUTER SCIENCE

Background

Self-driving car

- **Structured** environments
- **Traffic rules**
- Interaction with **homogeneous** agents
- **Safety**-critical system



Delivery robot

- **Unstructured** environments
- **Social rules**
- Interaction with **heterogenous** agents
- **Mission**-critical system



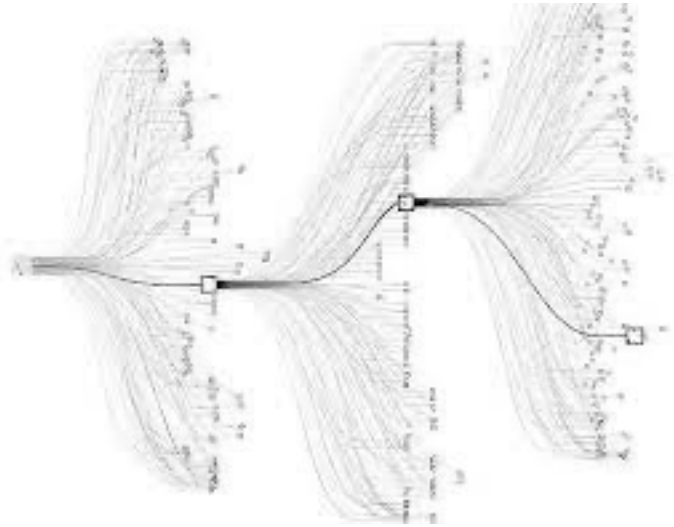
Human-like navigation

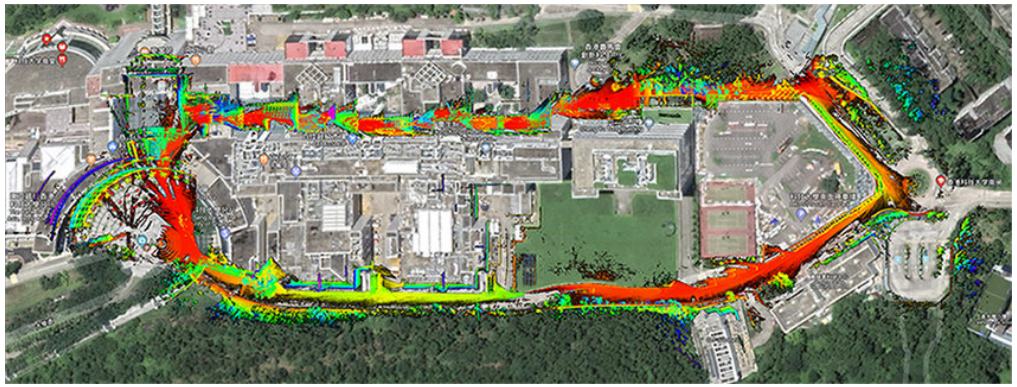
Humans are good at:

- **Fuzzy** reasoning
- **Semantics** abstract
- **Local** positioning
- **Online** scene understanding

Robots are good at:

- **Model-based** searching
- **Spatial** reconstruction
- **Global** positioning
- **Offline** HD Map





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Re-investigate our task:

What capabilities required in "navigation in human-living environments"?

Let's get some idea from human

Human's core skill:

decentralized collision avoidance



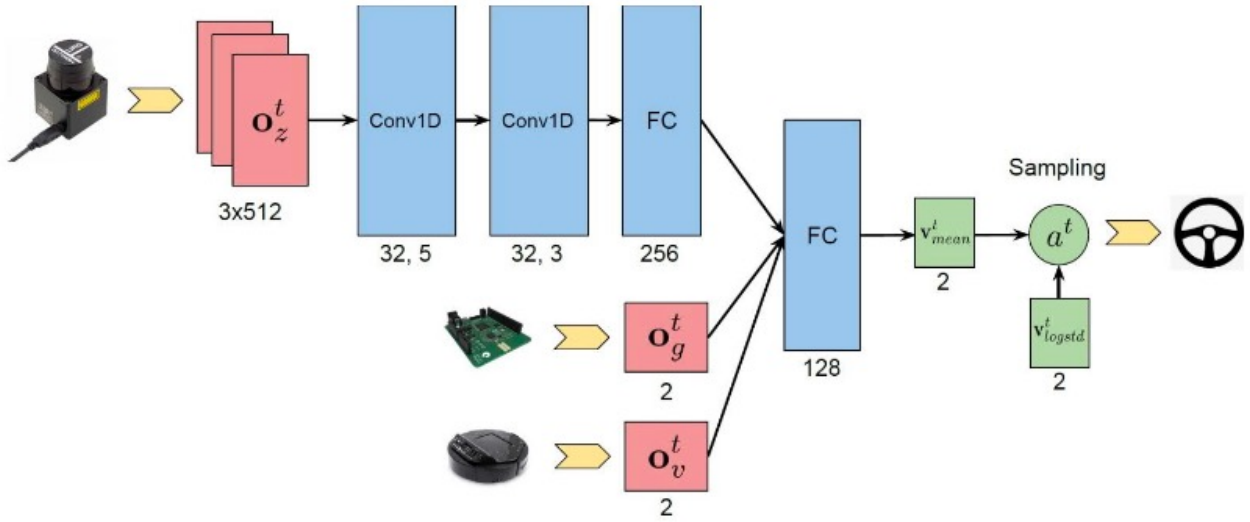
Human-like navigation in human-living environments



- Decentralized collision avoidance behavior (planning core)

Learn collision avoidance behavior via RL

- Sensor-based neural network controller

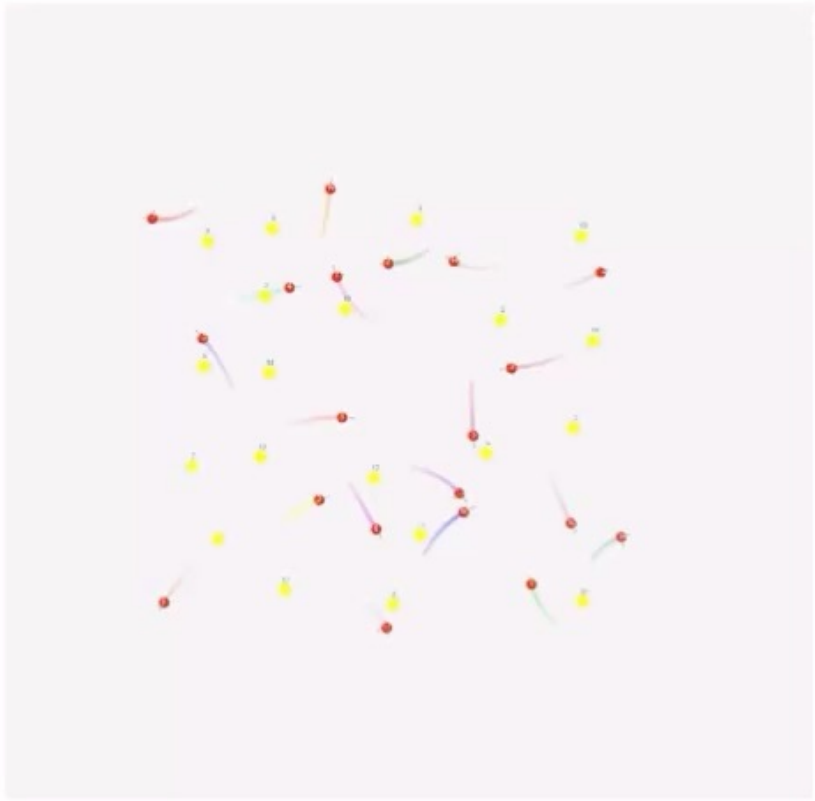


- Optimize the entire crowd for:
  - * Minimizing collision cost
  - * Minimizing the time to reach goal
  - * Maximizing trajectory smoothness

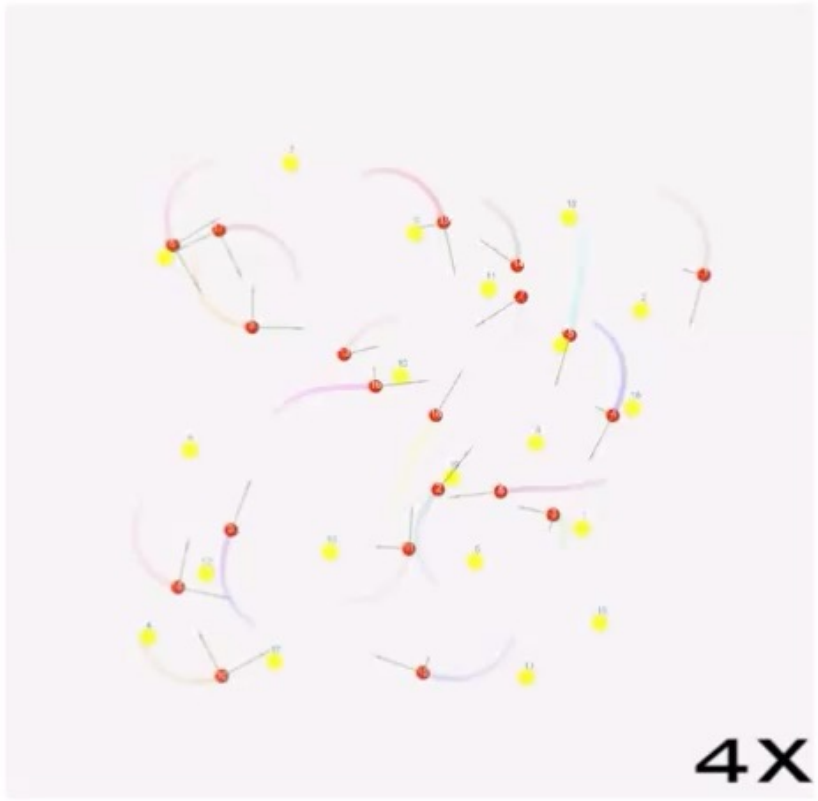
Encourage smooth cooperation behavior

Training step 1

- “Baby” controller trained on simple scenarios
- Random configuration for start and goal (yellow)



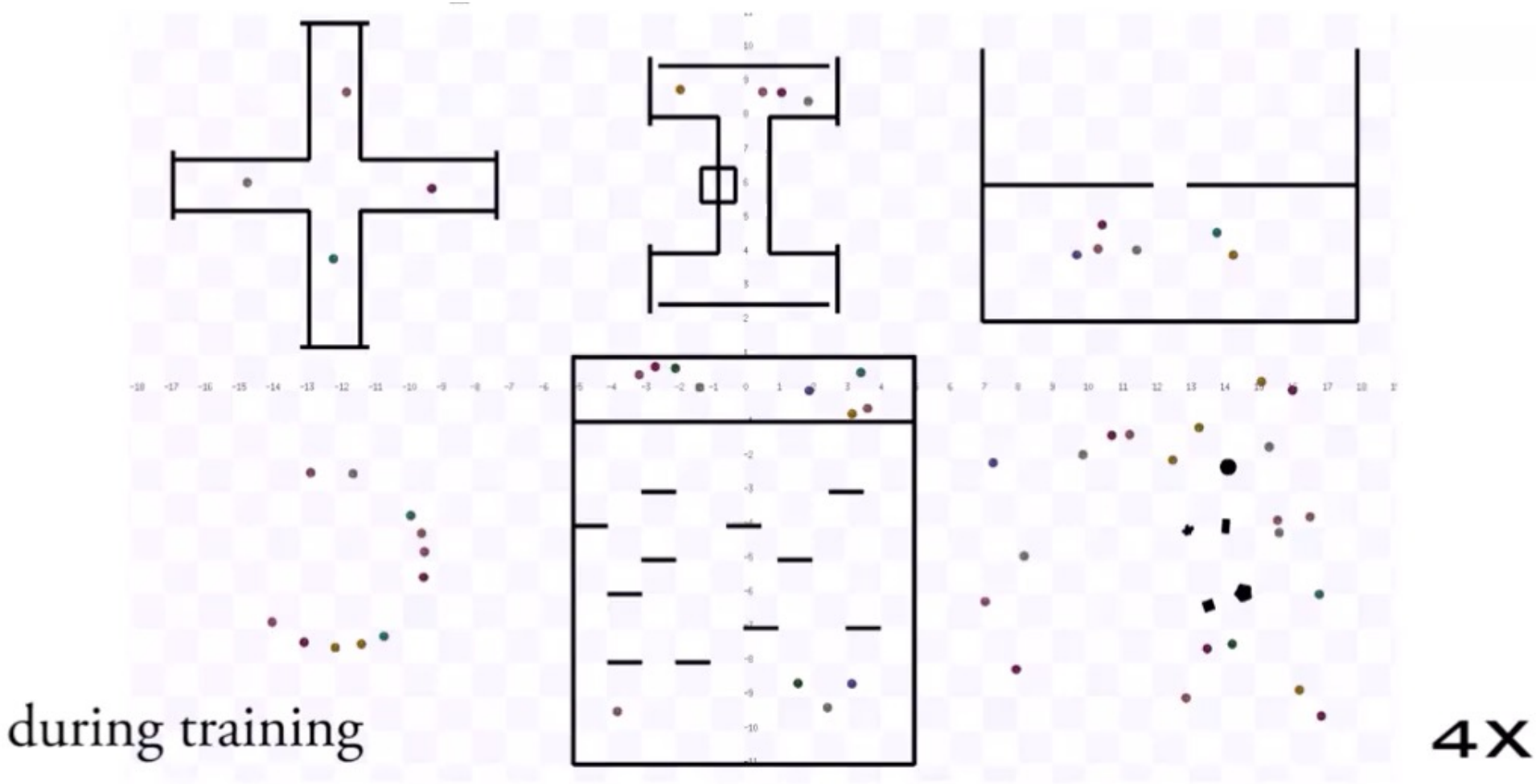
during training



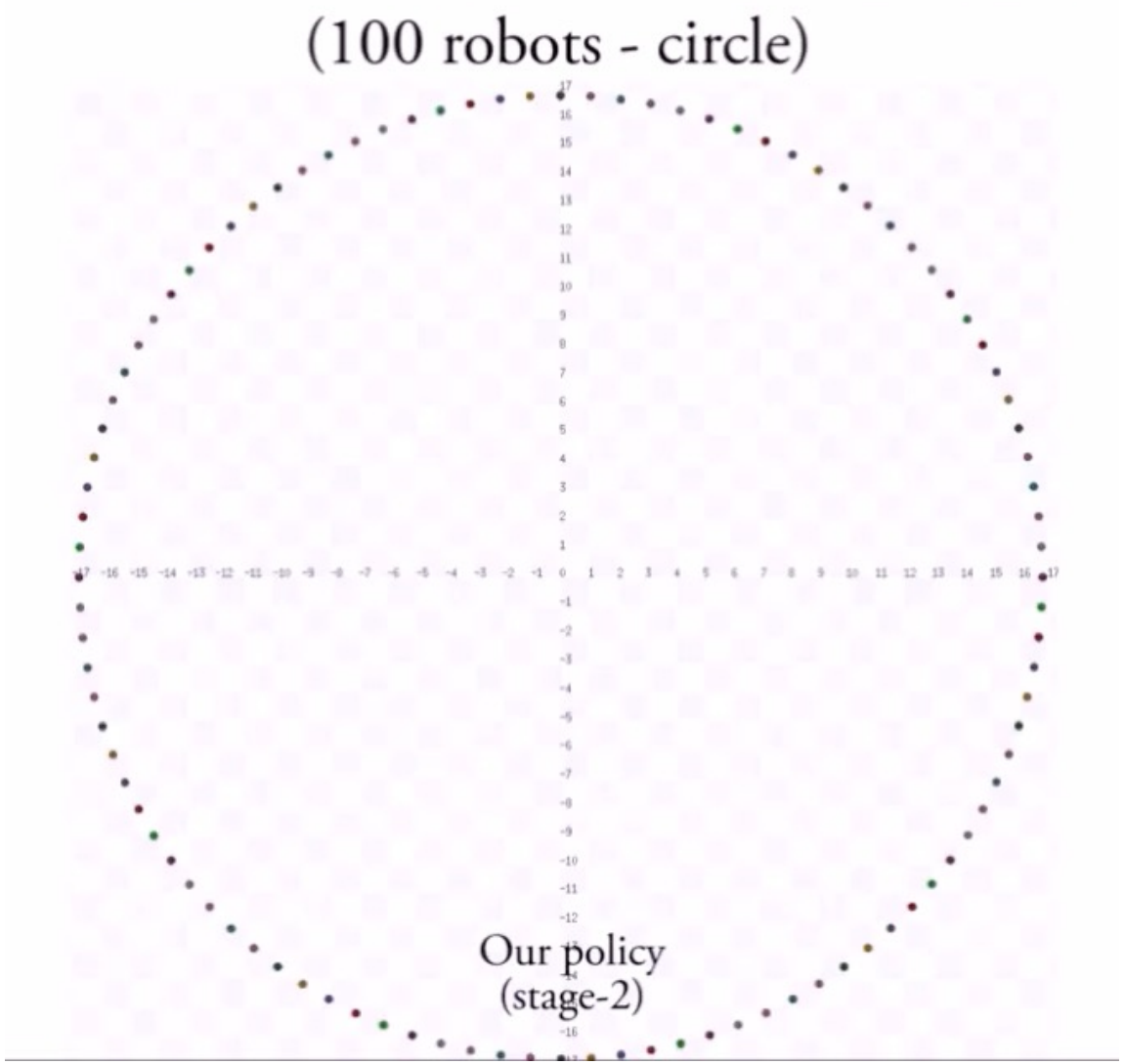
training finished

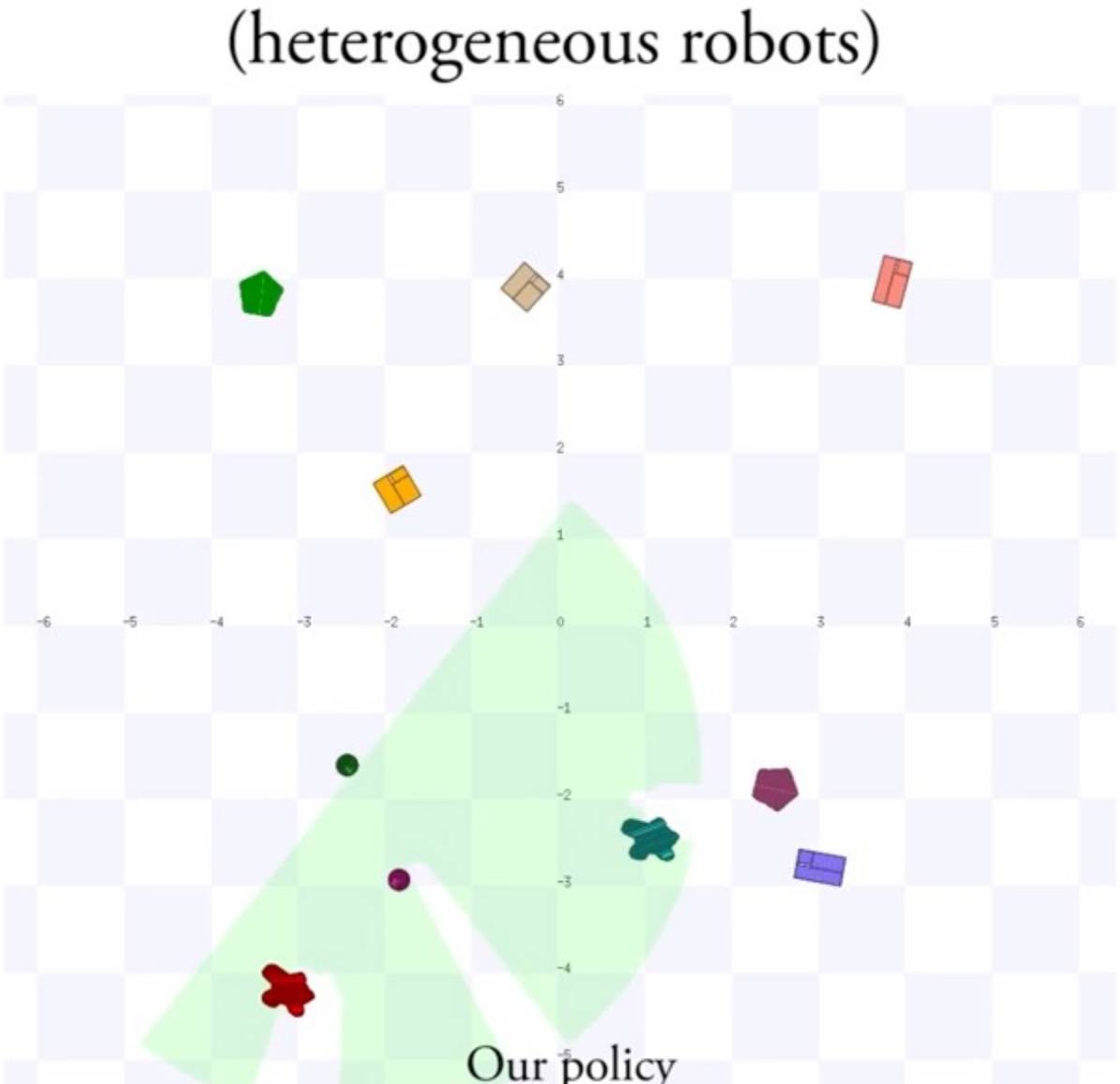
Training step 2

- Controller gets “mature” after being trained on multiple challenging scenarios



Test in a challenging benchmark





Real robot demo: decentralized multi-robot collision avoidance



Real robot demo: Navigation in Crowds



Real robot demo:
Sim-to-Real to different platforms



Real robot demo:
Sim-to-Real to different platforms



Human-like navigation in human-living environments



- **Decentralized collision avoidance behavior (Planning core)**
- Perception-aware decision making (Perception-assist Planning)

Complex trade-off between safety and efficiency

The cat is **dynamic** and of **high uncertainty**



The robot is too close to the cat

Corridor is **static** and of **low uncertainty**



The robot gets stuck at the corridor

How to learn a **resilient navigation** policy to **adapt to different environments** automatically.

Resilient Policy



Idea: establish a **connection** between the **environmental uncertainty** and the **action distribution**.

We can roughly plot the reward distribution of the robot in the lower part



Resilient Policy

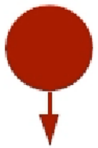
Idea: establish a **connection** between the **environmental uncertainty** and the **action distribution**.

The environmental uncertainty will respond to the variance of the action distribution by the connection.

Resilient Policy

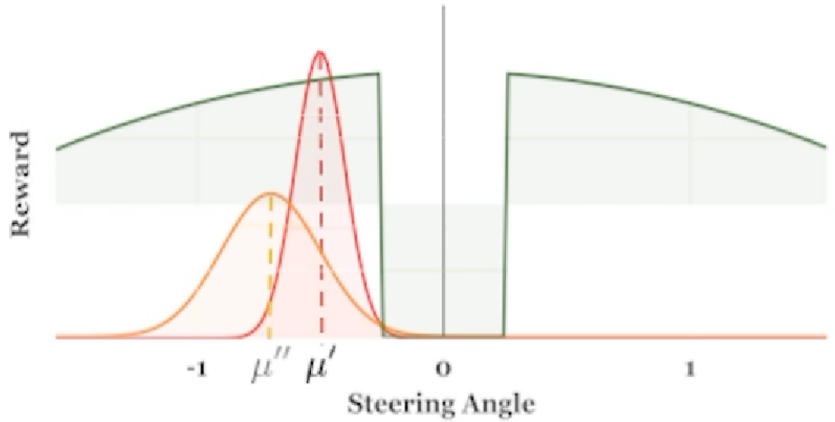
Idea: establish a **connection** between the **environmental uncertainty** and the **action distribution**.





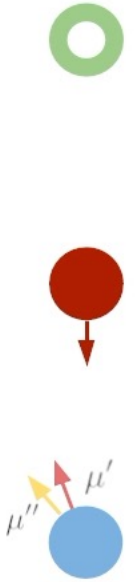


To maximize the reward, the robot can adaptively generate the mean value of the action distribution.

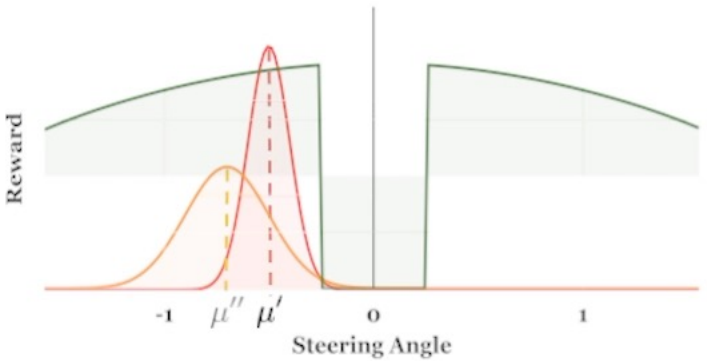


Resilient Policy

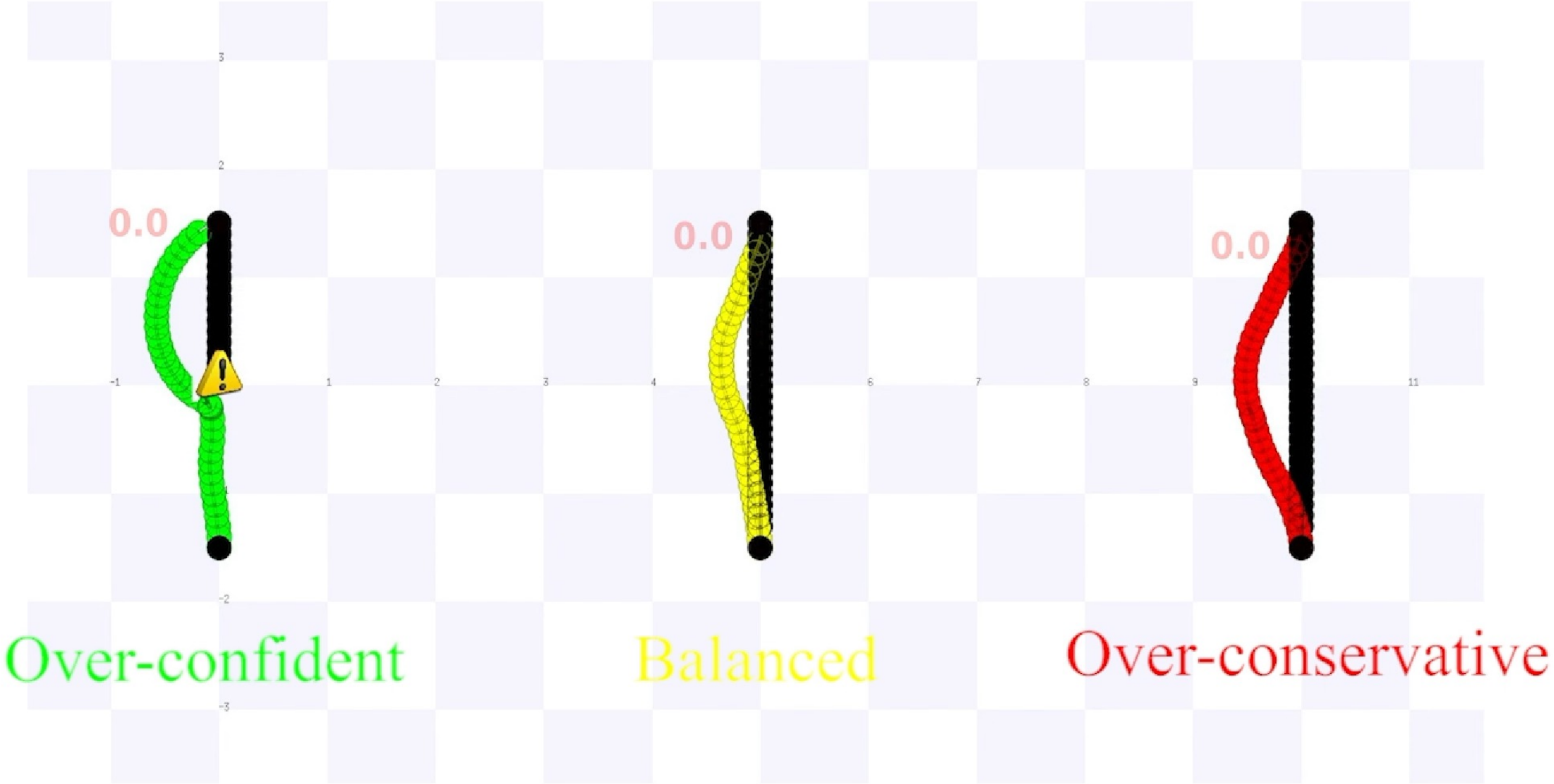
Idea: establish a **connection** between the **environmental uncertainty** and the **action distribution**.



Resilient behavior is learned by the robot: **more conservative** when facing **higher environmental uncertainty**

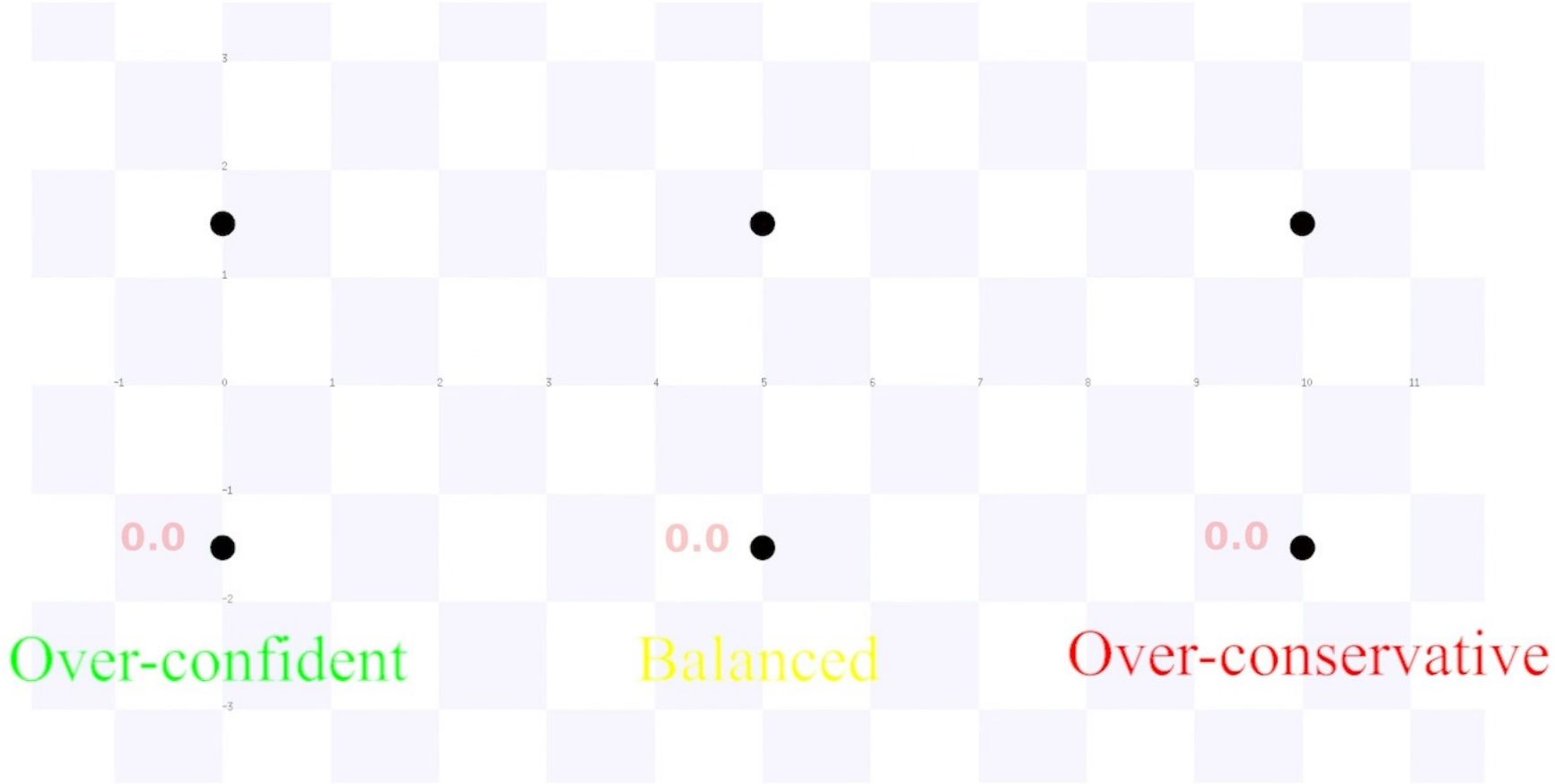


Encounter Scenario



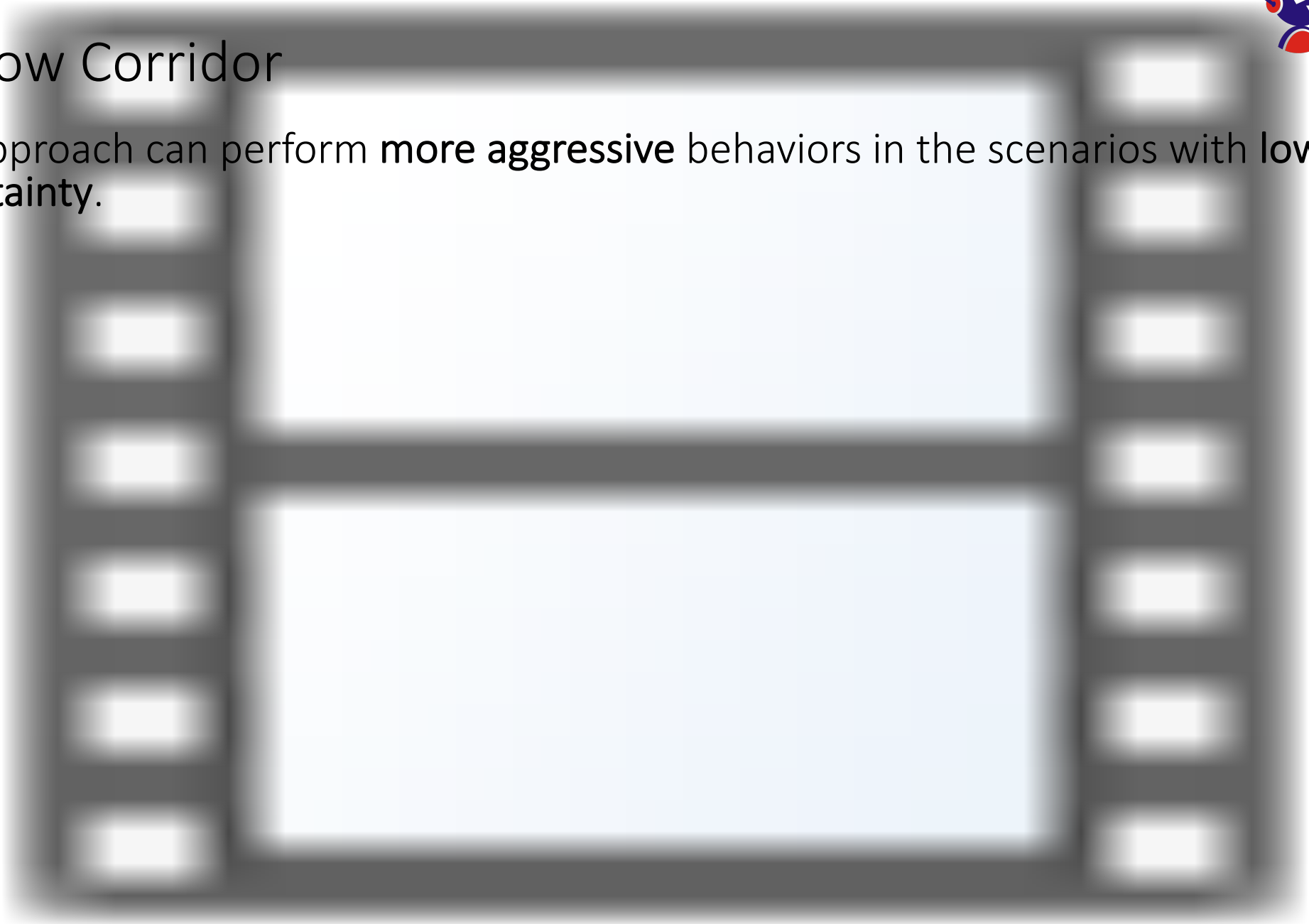
Encounter Scenario

More conservative behaviors in environments with **higher uncertainty**.



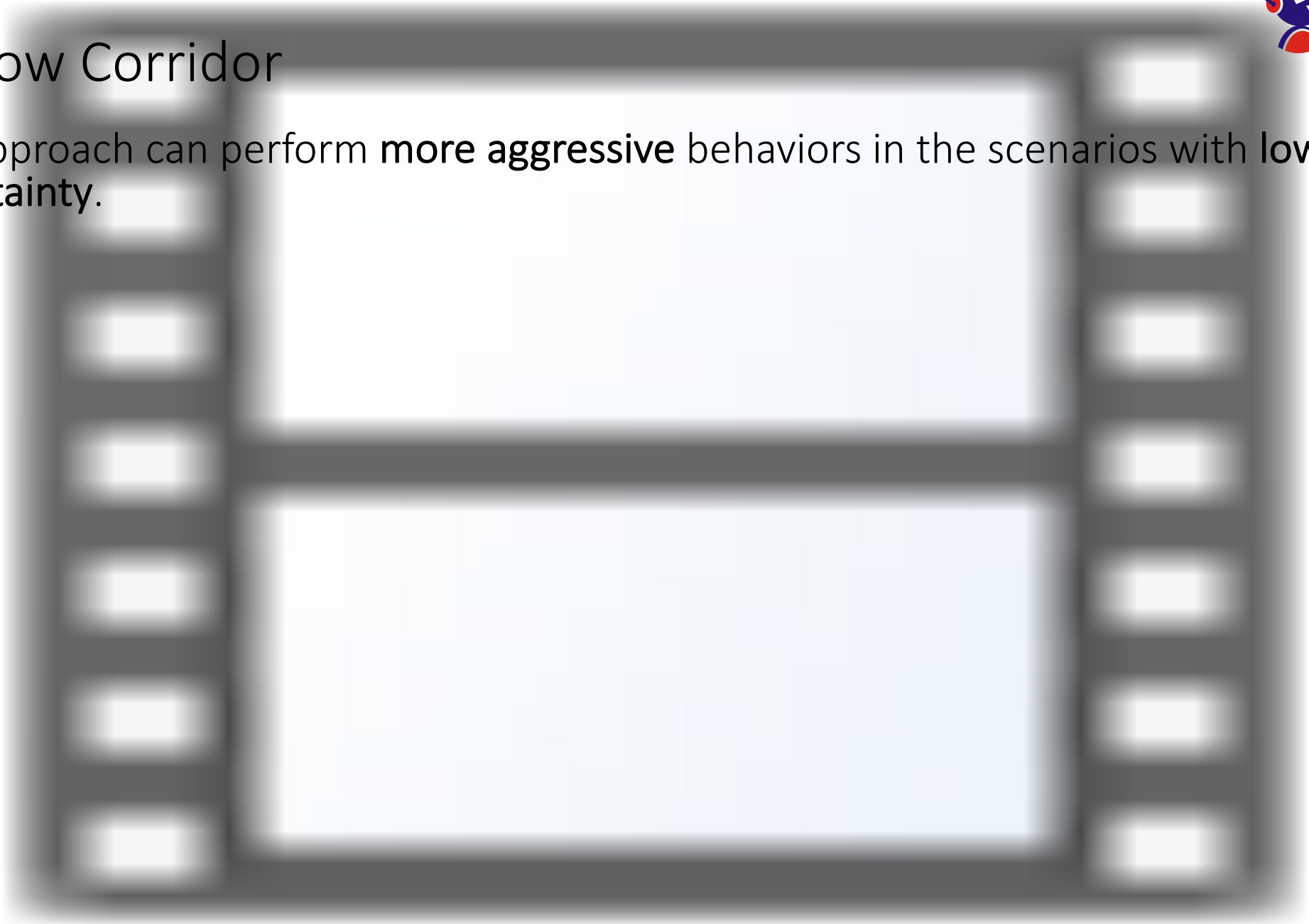
Narrow Corridor

Our approach can perform **more aggressive** behaviors in the scenarios with **lower uncertainty**.



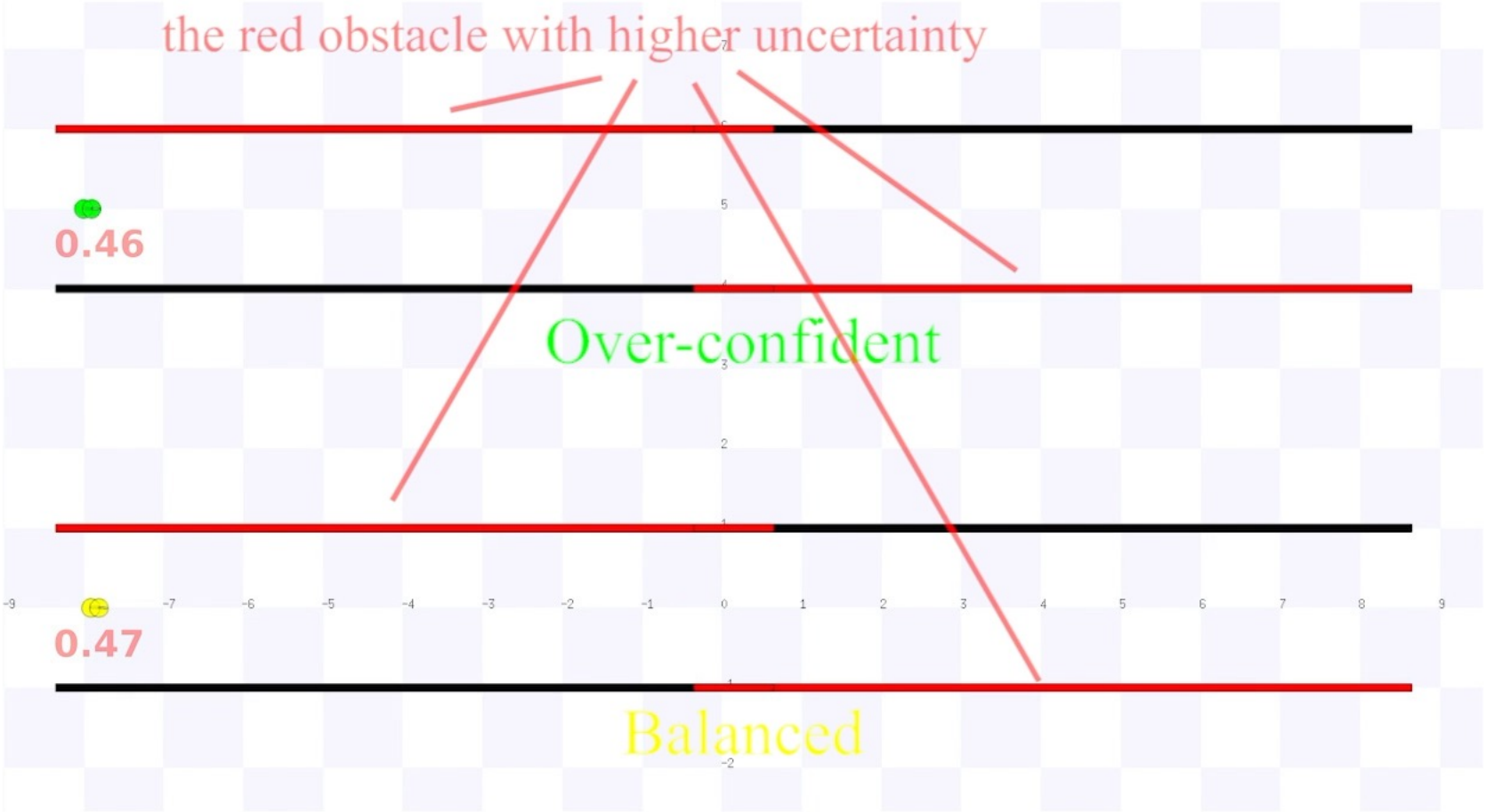
Narrow Corridor

Our approach can perform **more aggressive** behaviors in the scenarios with **lower uncertainty**.



Uncertain Corridor

In this experiment, we further investigate whether our approach has learned **uncertainty-averse behavior**.



Human-like navigation in human-living environments



- **Decentralized collision avoidance behavior (Planning core)**
- **Perception-aware decision making (Perception-assist planning)**
- Recovery localization in crowds (Planning-assist perception)

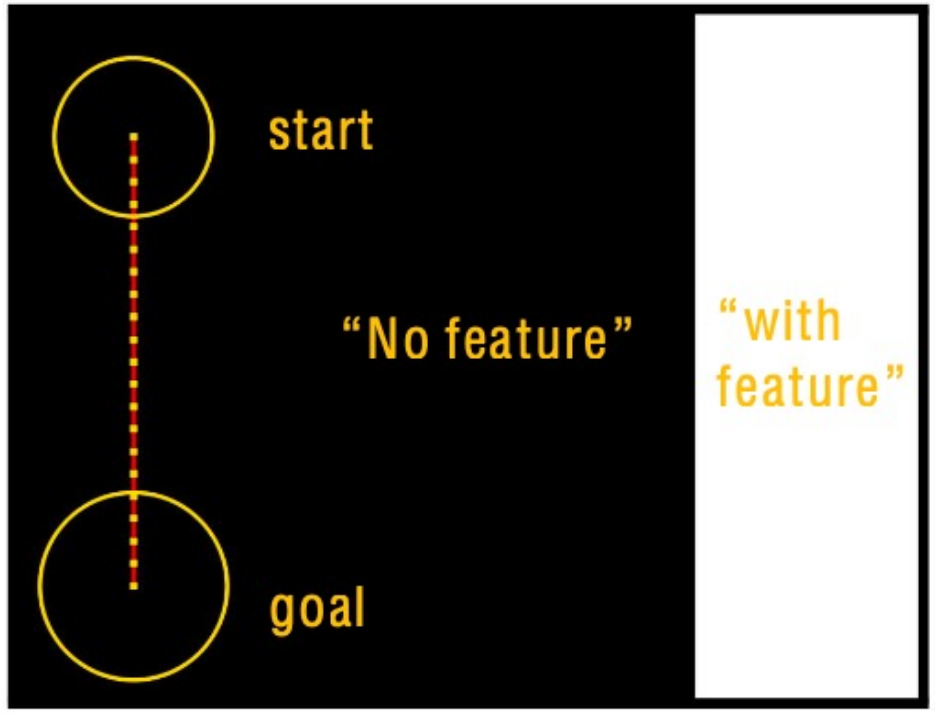
Traditional localization in dense crowds

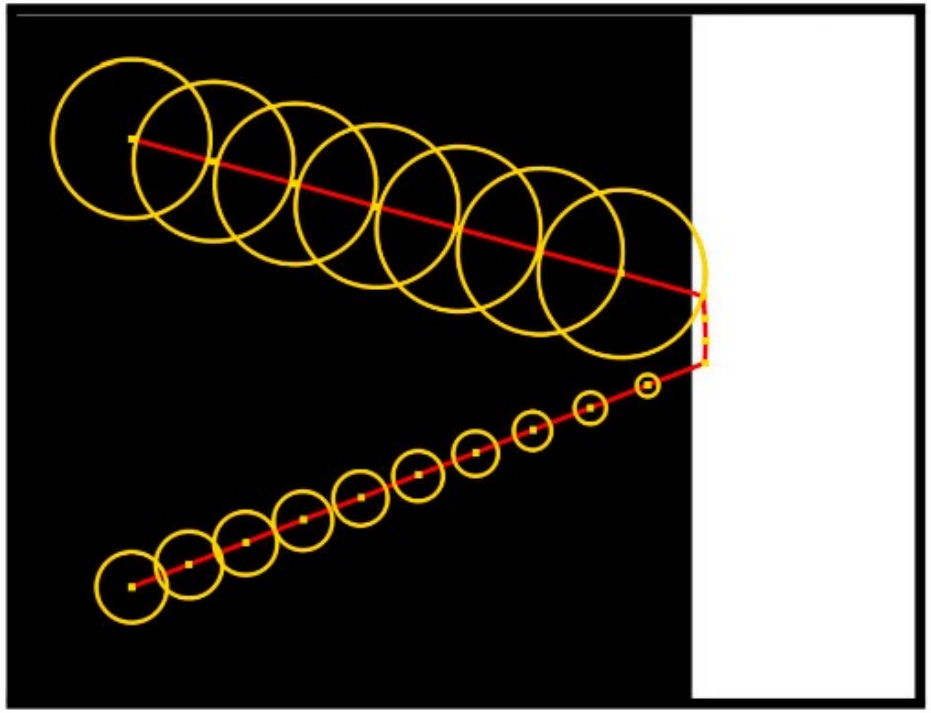
- Need to overcome the static feature lost difficulty



Theoretical solutions

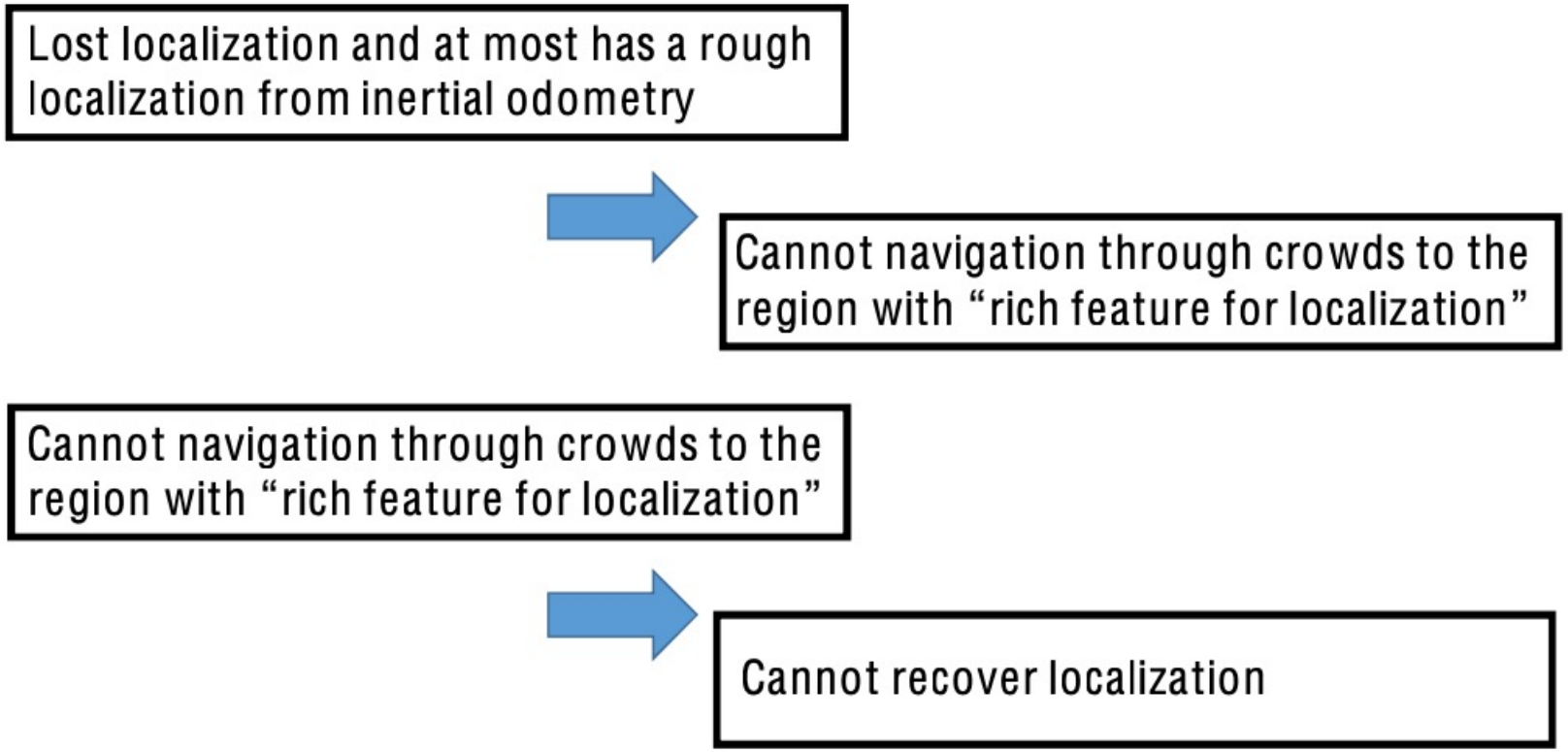
- Gather information by moving to regions with rich localization features, and then continue



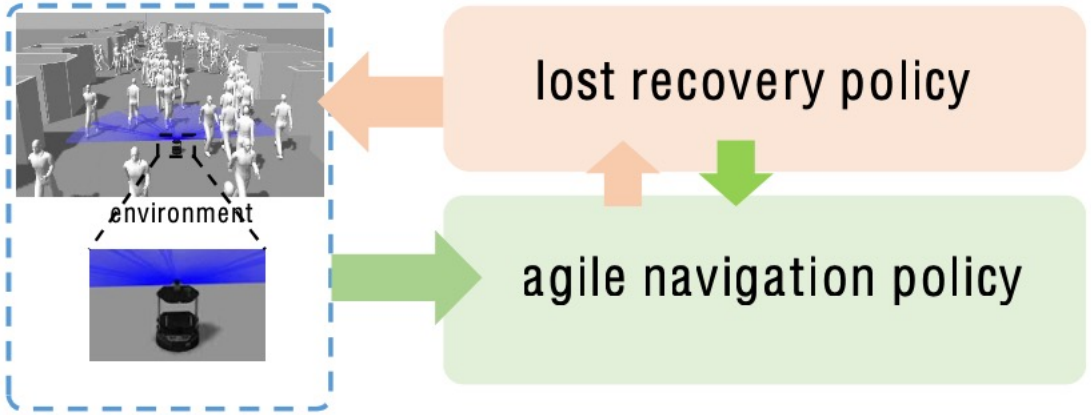


Chicken-and-egg difficulty

- But how to execute such “information gain” movement in a dense crowd?

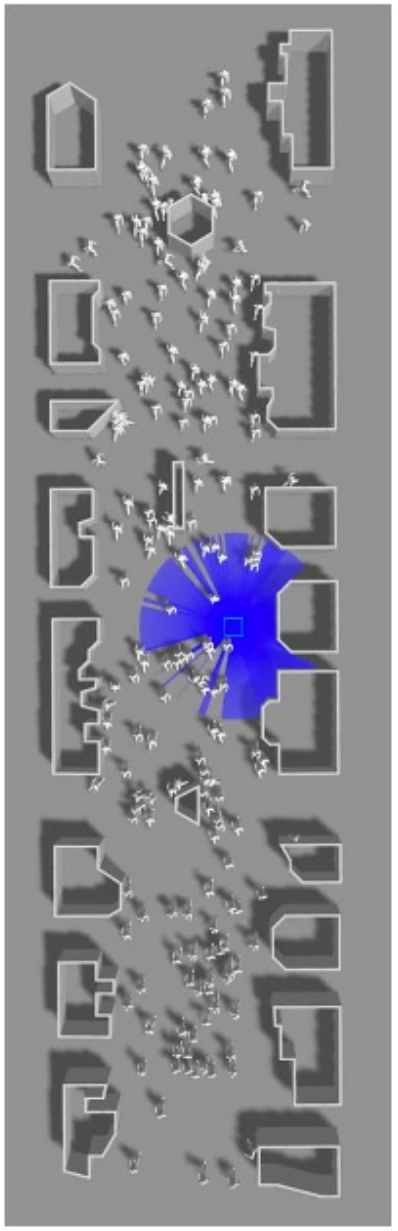


Our solution



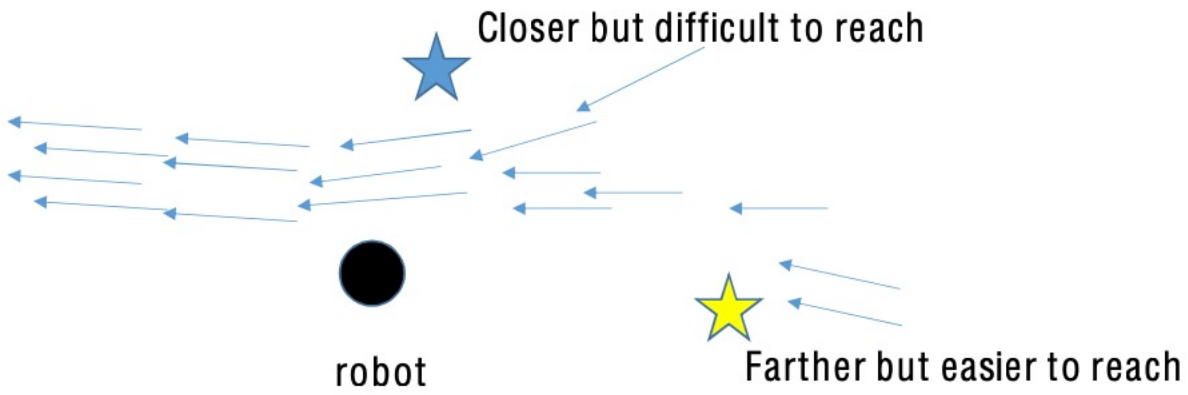
- Our collision avoidance does not rely on localization
- Our “moving to the goal” can safely use a rough localization to indicate where the feature-rich region.

Chicken-and-egg problem solved

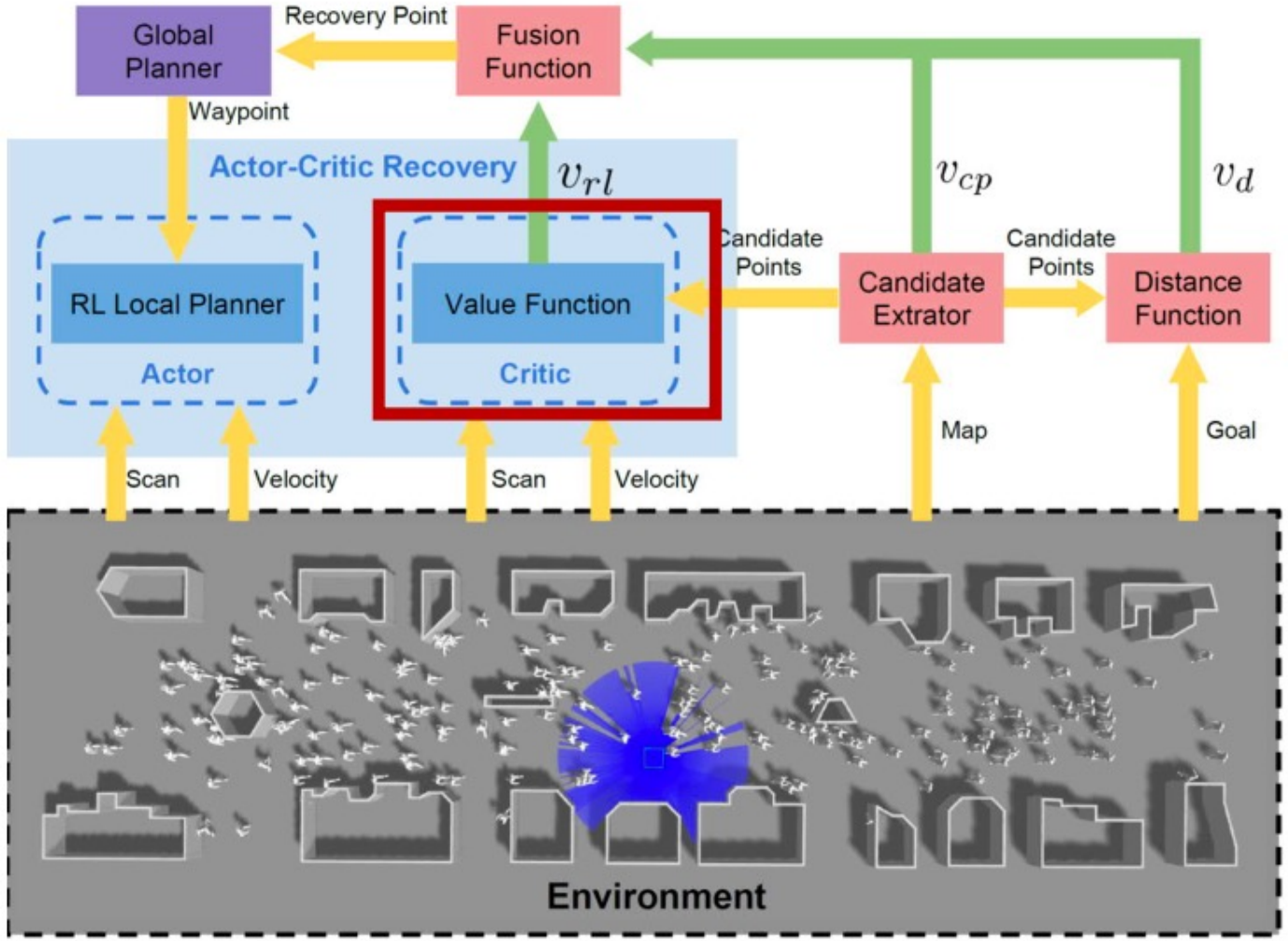


Localization recovery policy

- Choose a region in the map as a temporary goal for observing features and recover localization.
- Trade-off different regions according to:
  * The distance to the robot
  * The richness of features
  * How difficult for the robot to pass through the crowd flow and reach the region



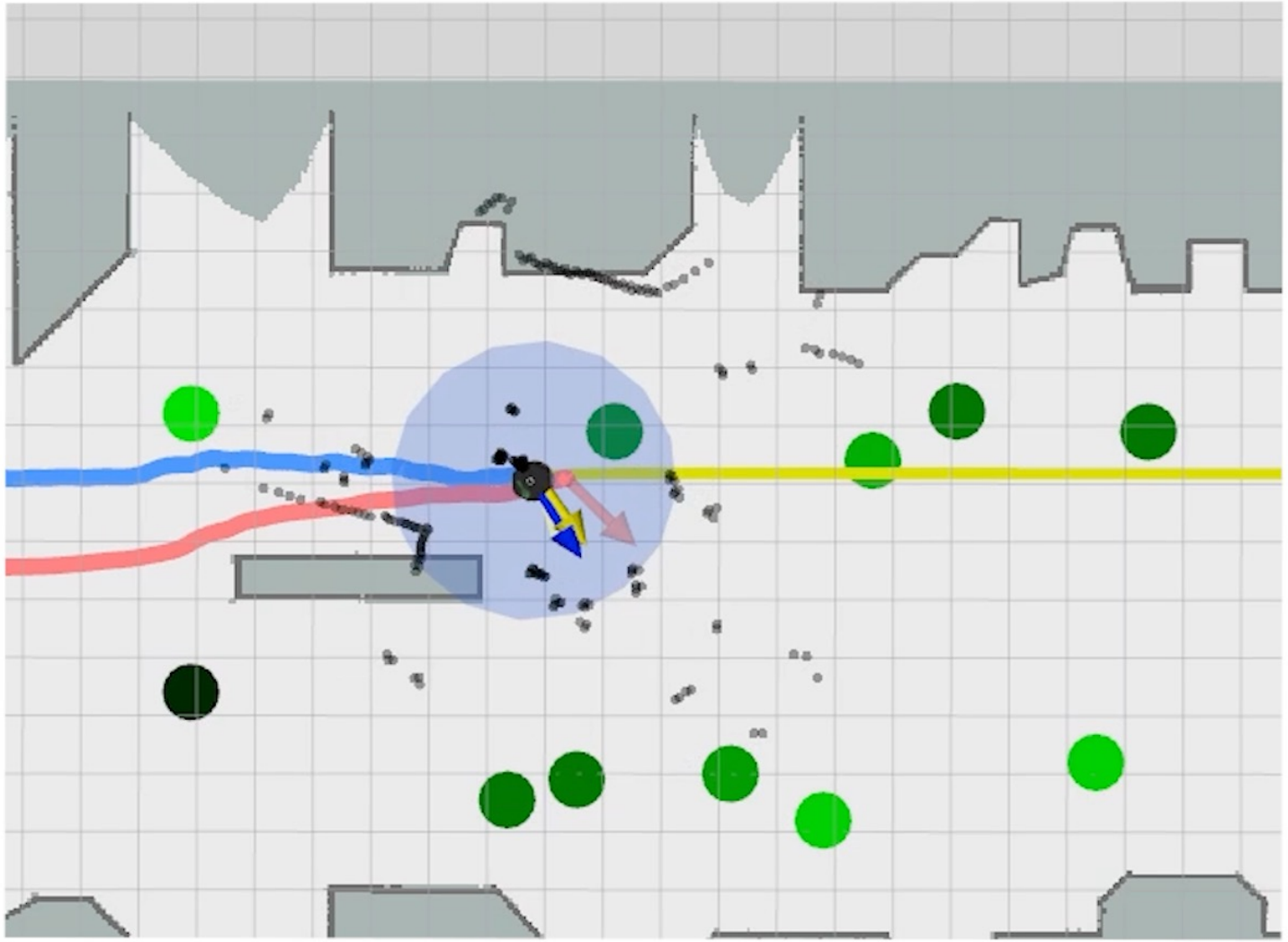
Actor-Critic recovery



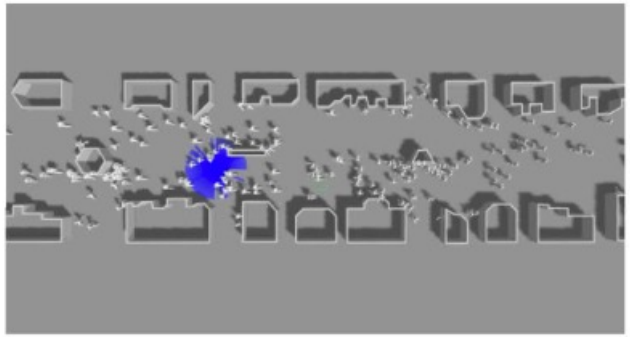
- The crowd-related criterion can be computed using the **value function**, a by-product of collision avoidance policy training
- Adaptively update policy according to the crowd-flow status

Recovery policy demo

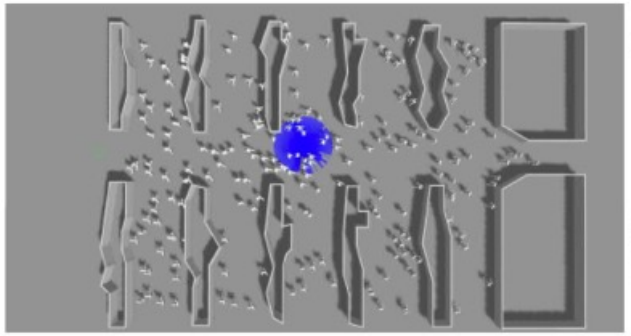
* Adaptive plan the trajectory to avoid the crowds



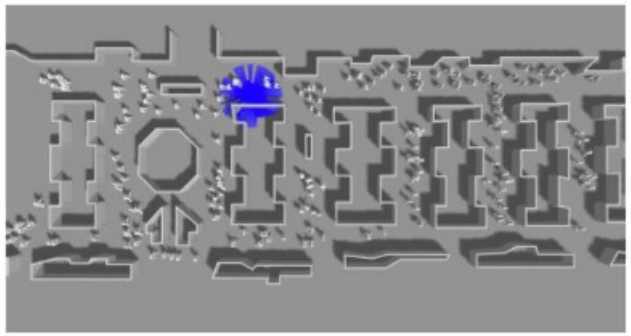
Simulation benchmarks for quantitative comparison



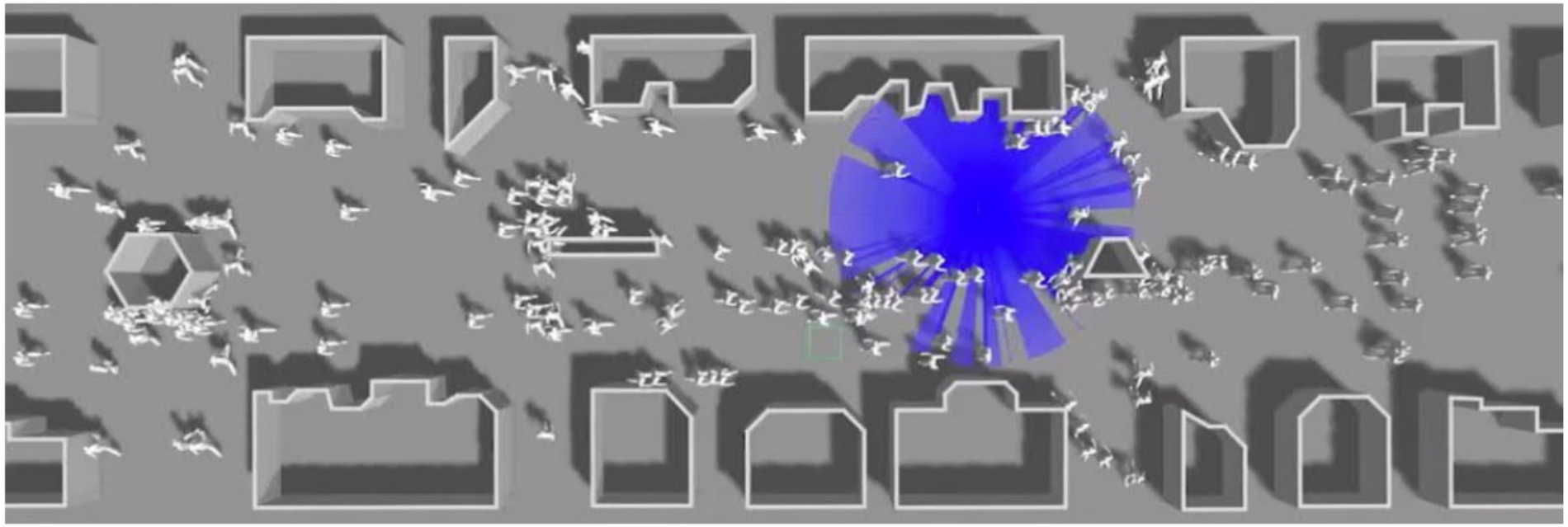
(a) corridor environment



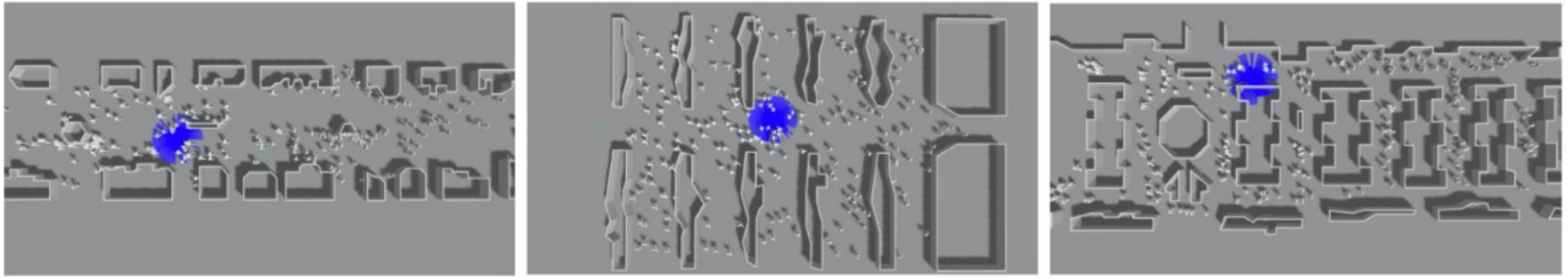
(b) supermarket environment



(c) airport environment



Simulation benchmarks for quantitative comparison



(a) corridor environment (b) supermarket environment (c) airport environment

Methods	Corridor Scenario				SuperMarket Scenario				Airport Scenario			
	Lost	Frozen	Success	Velocity	Lost	Frozen	Success	Velocity	Lost	Frozen	Success	Velocity
baseline	32%	65%	3%	0.52 (0.074)	18%	82%	0%	0	16%	84%	0%	0
RL	63%	17%	20%	0.76 (0.065)	86%	10%	4%	**0.81 (0.095)**	71%	5%	24%	**0.68 (0.051)**
RL-CP	44%	6%	50%	**0.81 (0.070)**	28%	8%	64%	0.64 (0.038)	33/%	31%	36%	0.51 (0.049)
$(RL)^2$	36%	3%	**61%**	0.73 (0.079)	20%	12%	**68%**	0.78 (0.089)	10%	28%	**62%**	0.60 (0.090)

Real-world demo:
Localization recovery demo



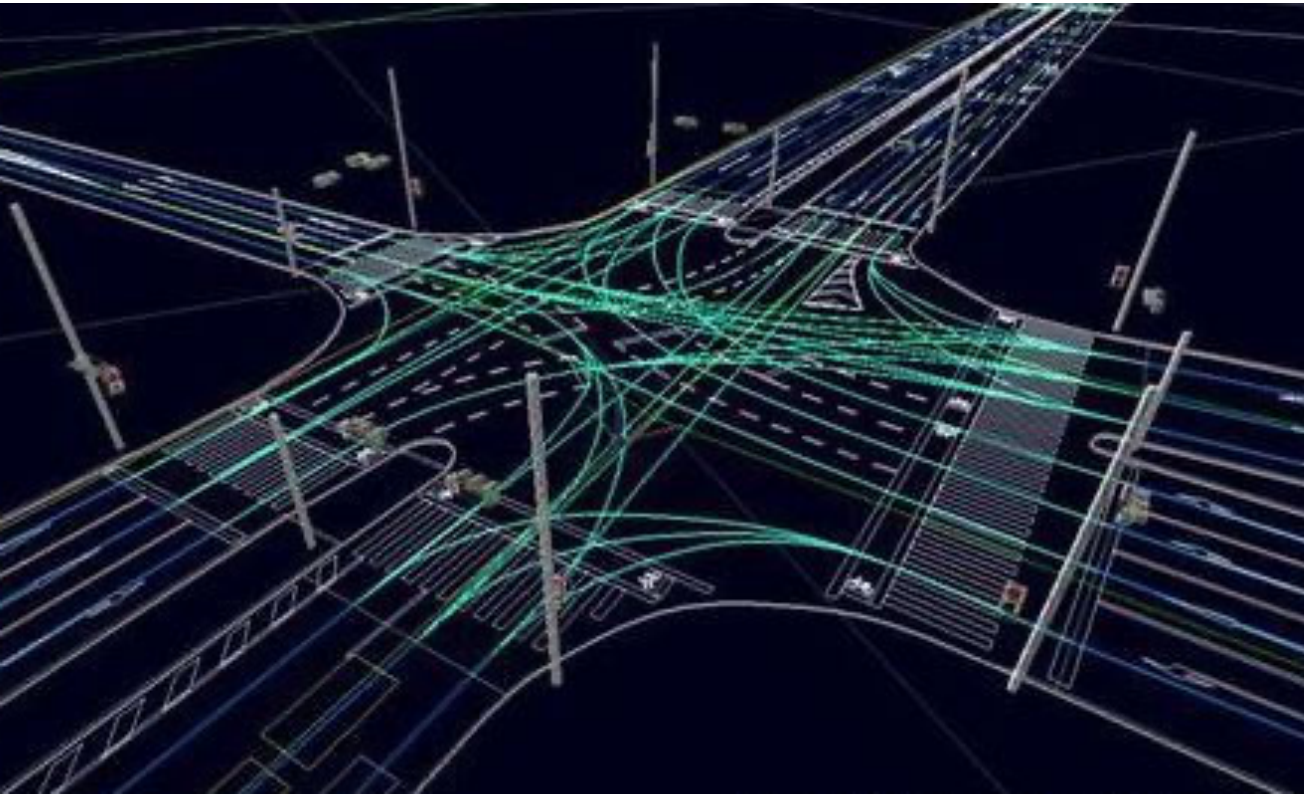
Human-like navigation in human-living environments

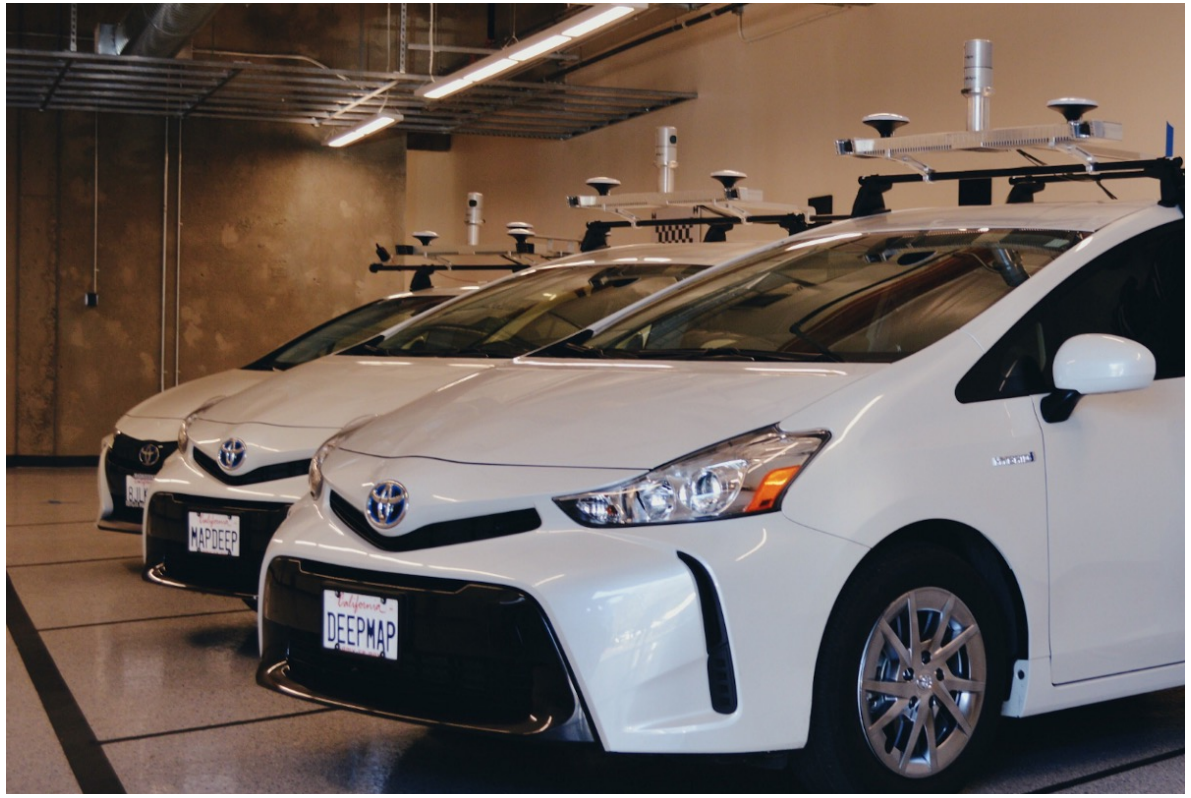


- **Decentralized collision avoidance behavior (Planning core)**
- **Perception-aware decision making (Perception-assist planning)**
- **Recovery localization from crowds (Planning-assist perception)**
- Online scene understanding (Perception core)

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Maintaining HD Map would be a tremendous effort.



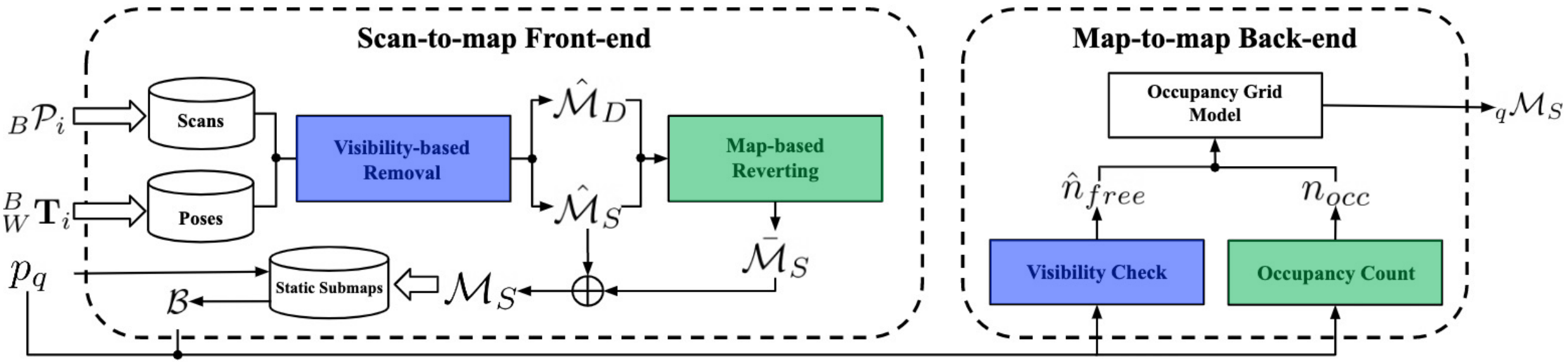


Online scene understanding for spatial structures. (static)

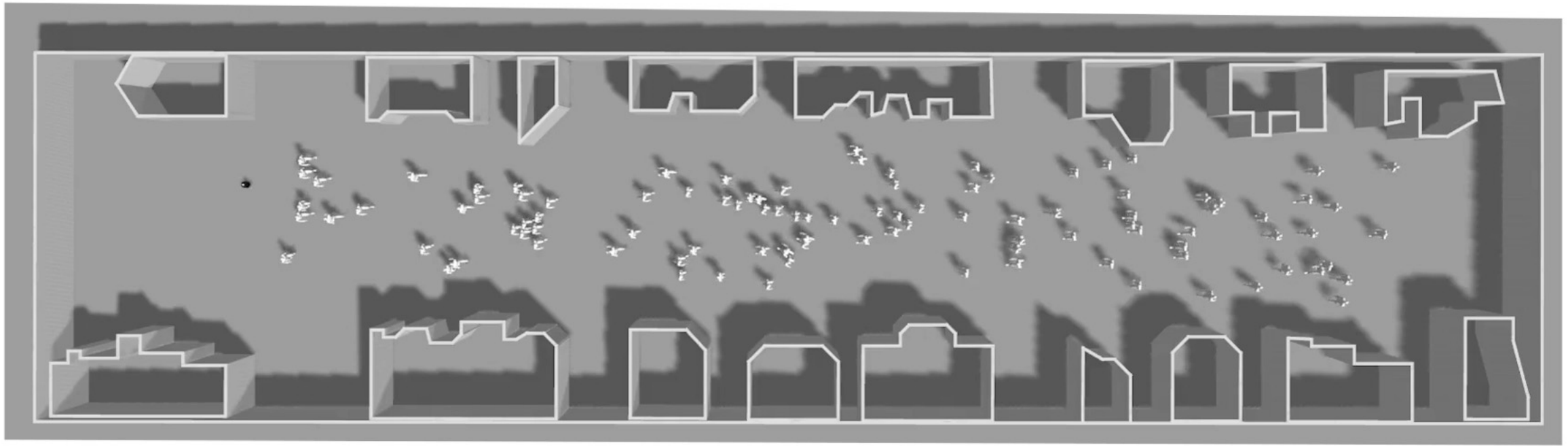
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Online dynamic objects removal (SLAM-like framework)



- Map-based approach: Slow, Good performance
- Visibility-based approach: Fast, Fair performance



To evaluate the performance of dynamic object removal, we introduce a simulation environment with many dynamic objects in Gazebo simulator.

Online scene understanding for dynamic agents.

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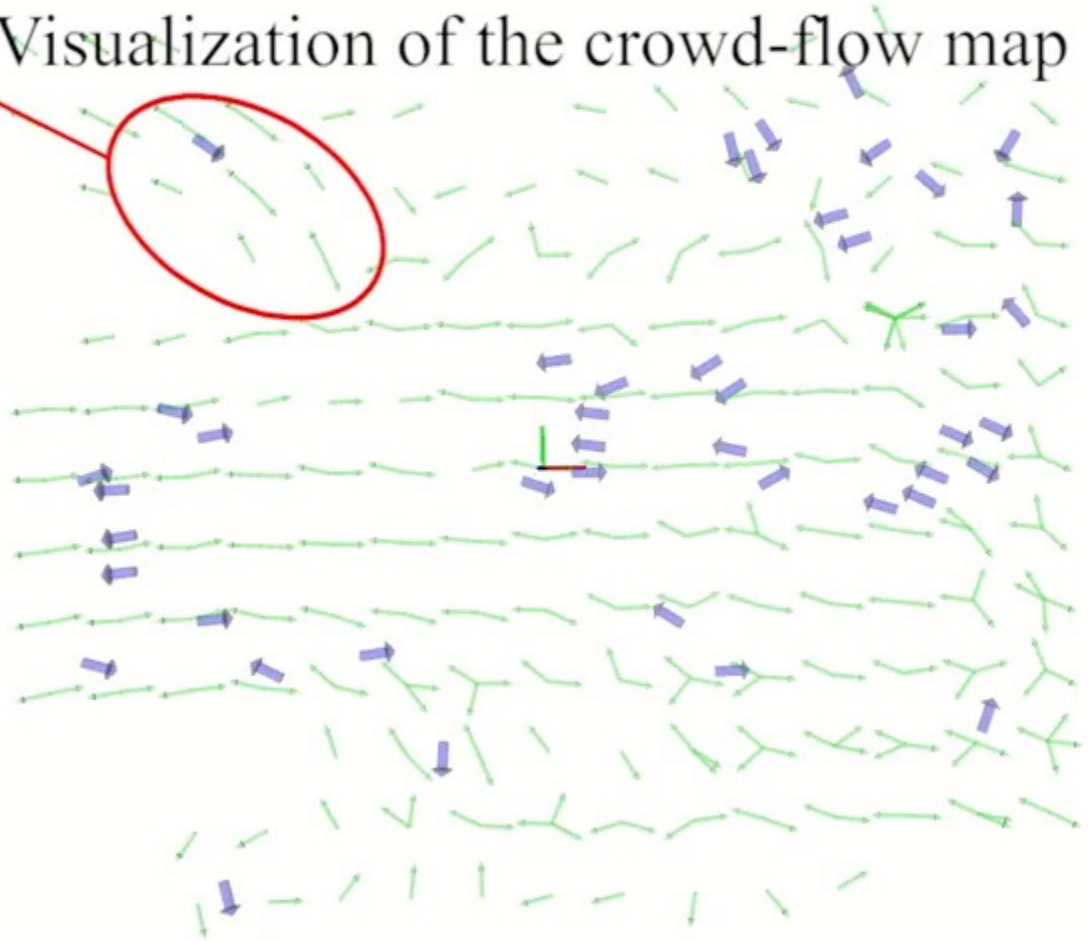


Motivation

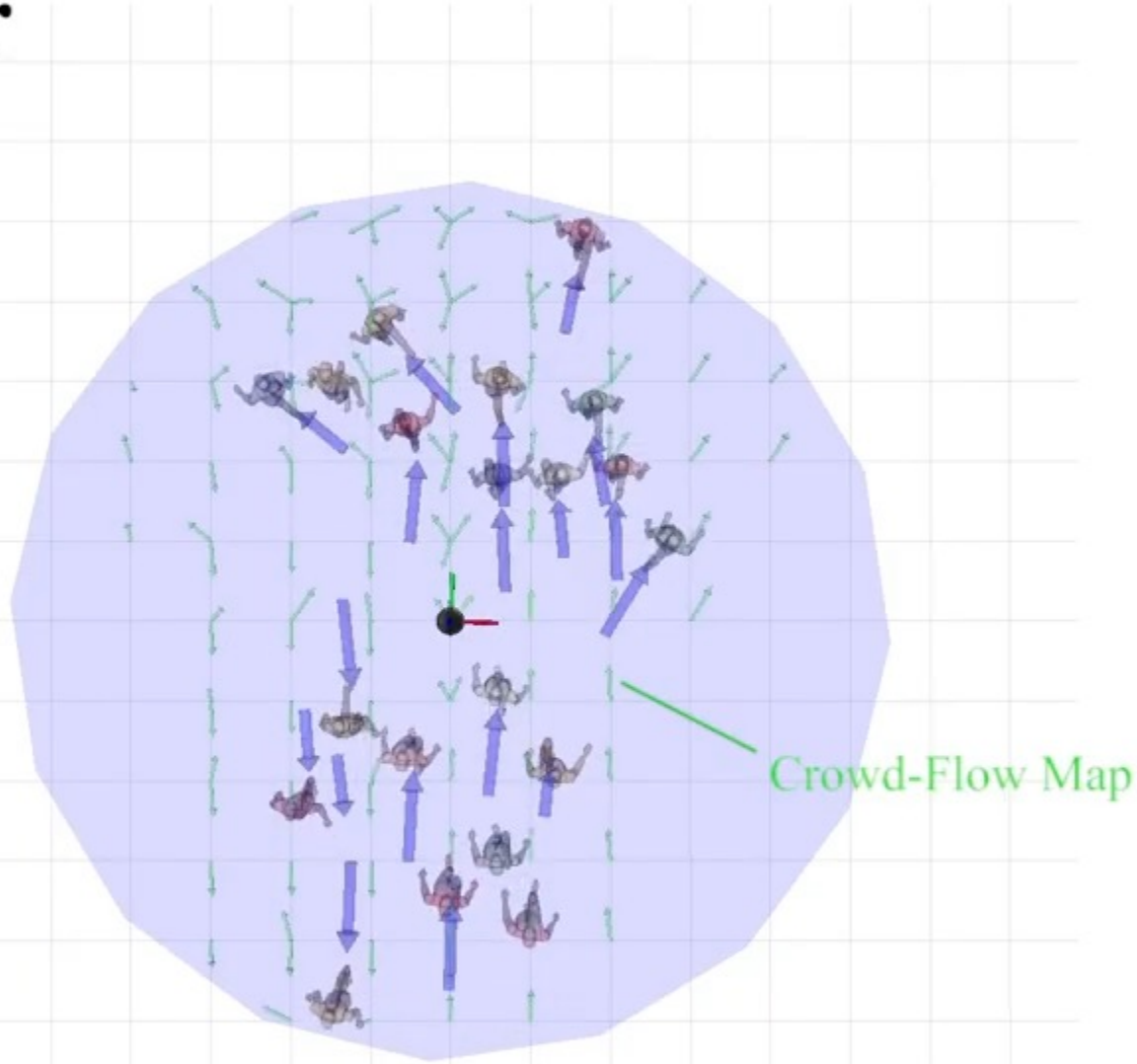
traversable area



Visualization of the crowd-flow map

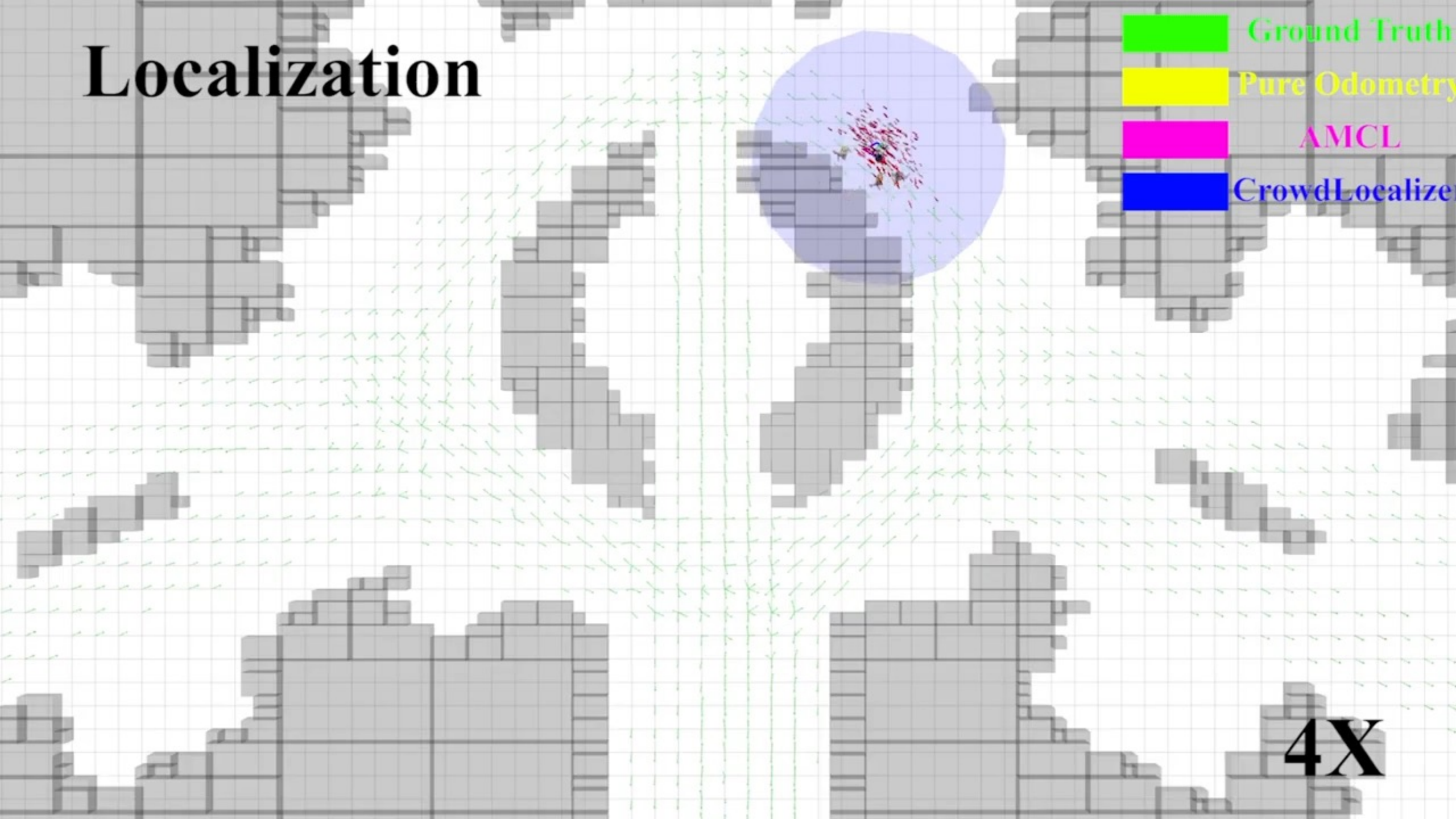


CrowdMapper



We apply clustering algorithm to model it into the crowd-flow map to represent the move pattern of pedestrians in the environment

Localization



Planning

A high-quality plan is expected to **minimize the inter-disturbance** with the crowd.

Thus we can use the number of the robot's active collision avoidance actions during the navigation to quantify whether the planned trajectory is socially compliant.

Robot Hardware



We mounted a 3D LiDAR and a multi-camera system on our robot to track the surrounding pedestrians.

Human-like navigation in human-living environments



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- **Recovery localization from crowds (Planning-assist perception)**
- **Online scene understanding (Perception core)**

Thanks for your listening